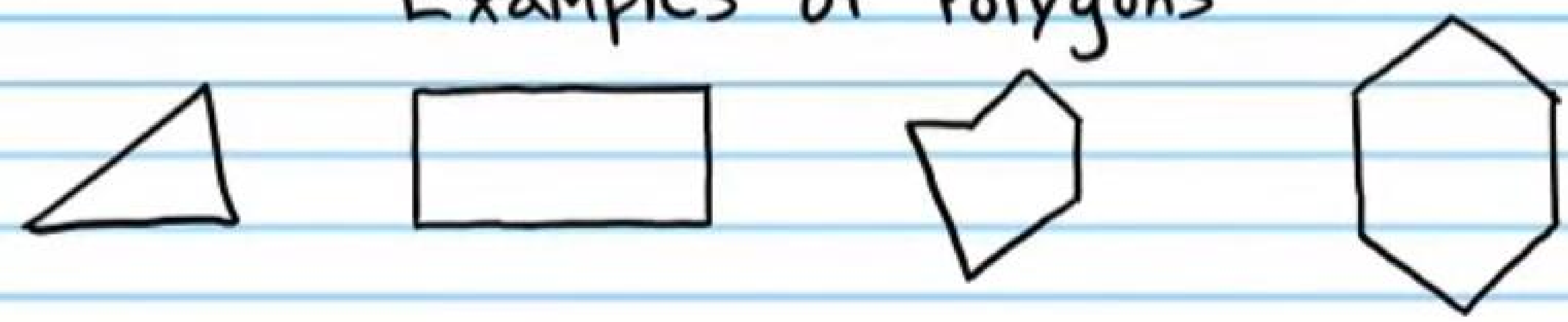
Name
Date
Course

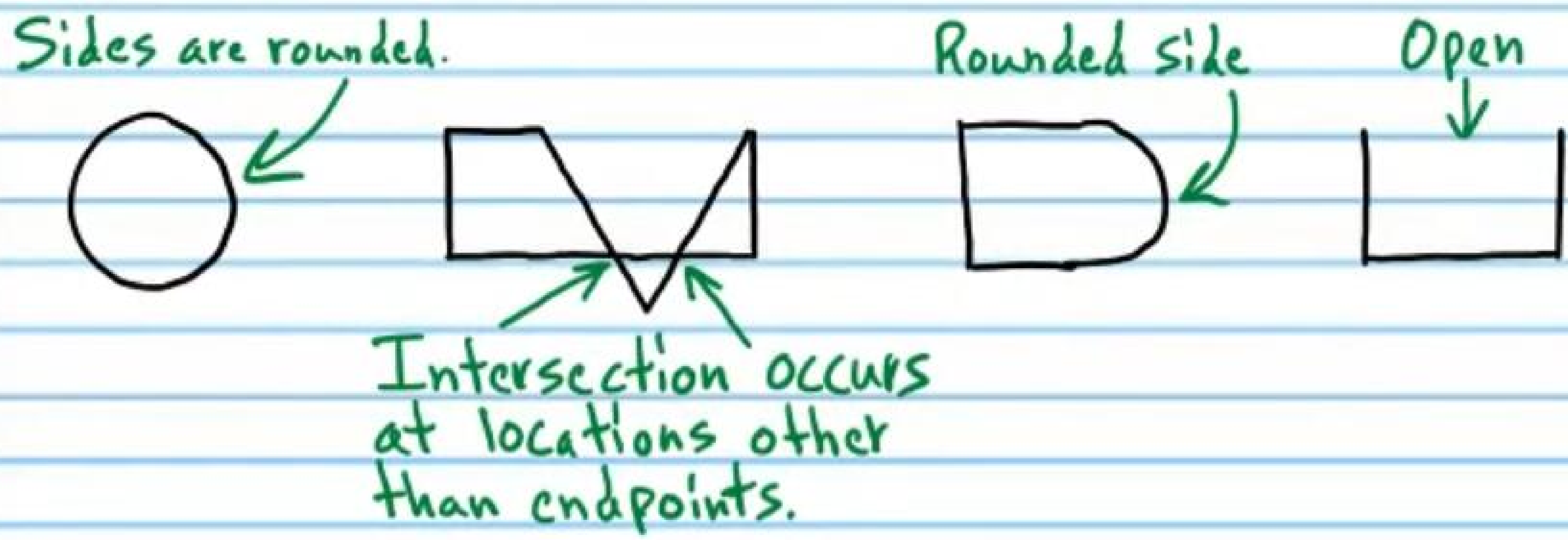
Fundamental Geometry (Part III)

Polygon: A closed plane figure whose sides are segments that intersect only at endpoints.

Examples of Polygons



Not Polygons

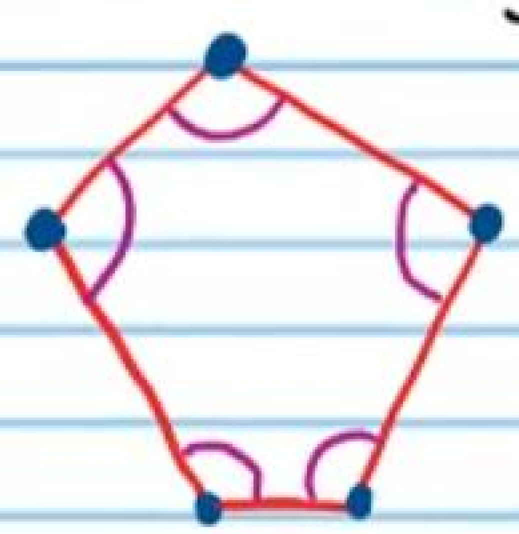


Parts of a Polygon

Vertices

Sides

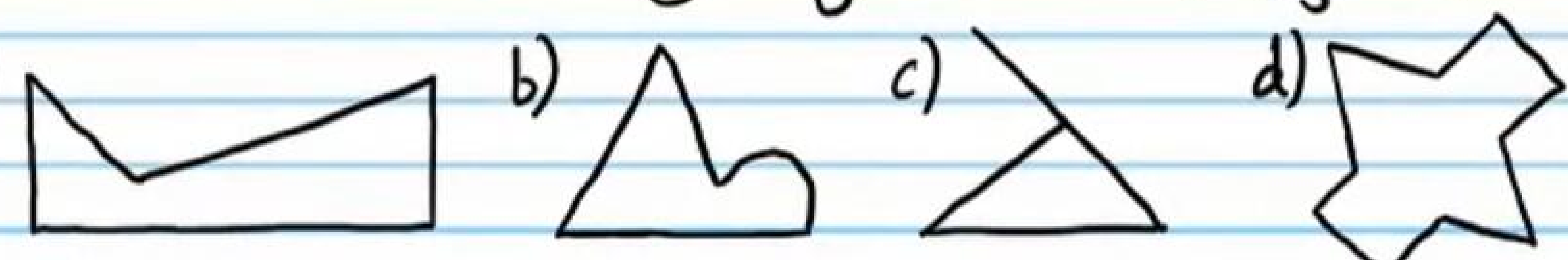
Interior Angles



Practice

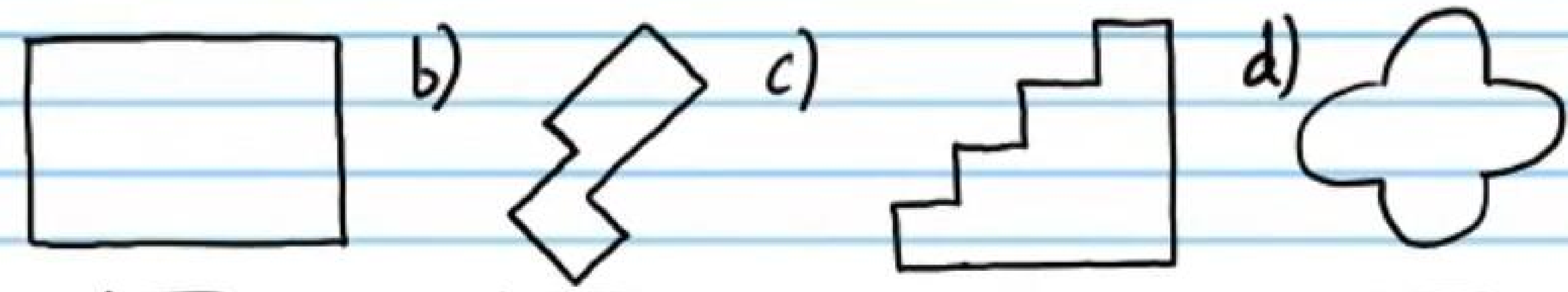
Which of the following figures are polygons?

1)



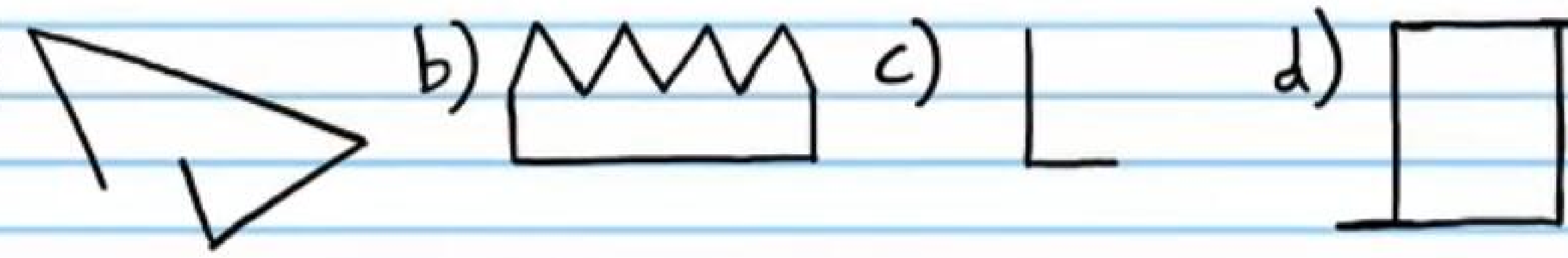
a) Yes b) No c) No d) Yes

2)



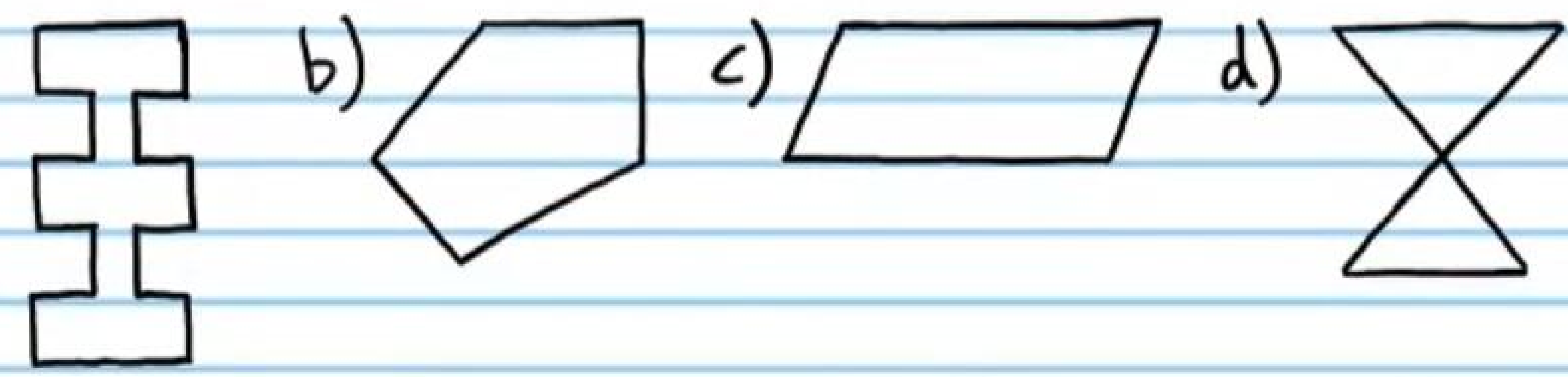
a) Yes b) Yes c) Yes d) No

3)



a) No b) Yes c) No d) No

4)



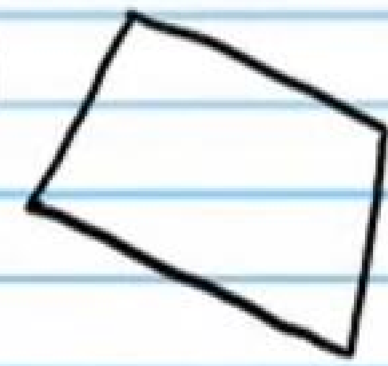
a) Yes b) Yes c) Yes d) No

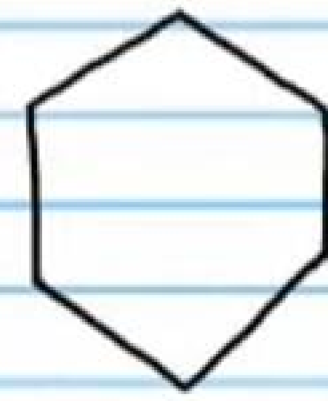
Names of Polygons

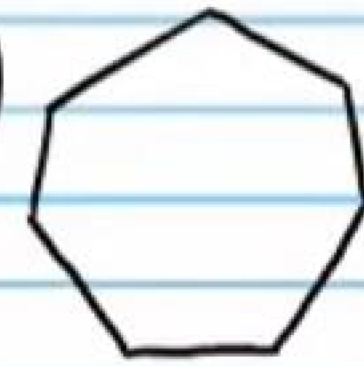
Sides	Name
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon (also called a Septagon)
8	Octagon
9	Nonagon
10	Decagon
12	Dodecagon
n	n-gon

Practice

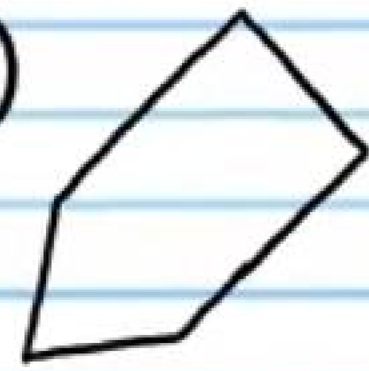
Name the following polygons.

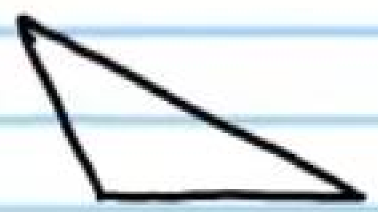
5 a)  Quadrilateral

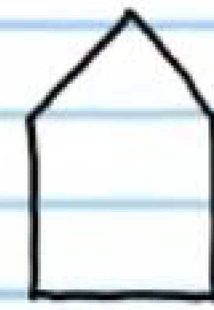
b)  Hexagon

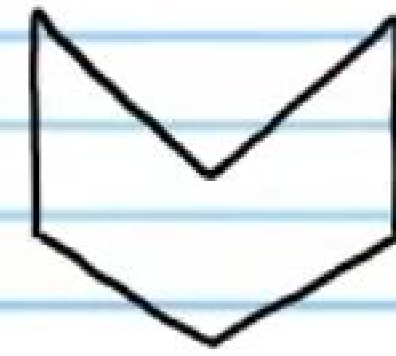
c)  Heptagon

d)  Octagon

6 a)  Pentagon

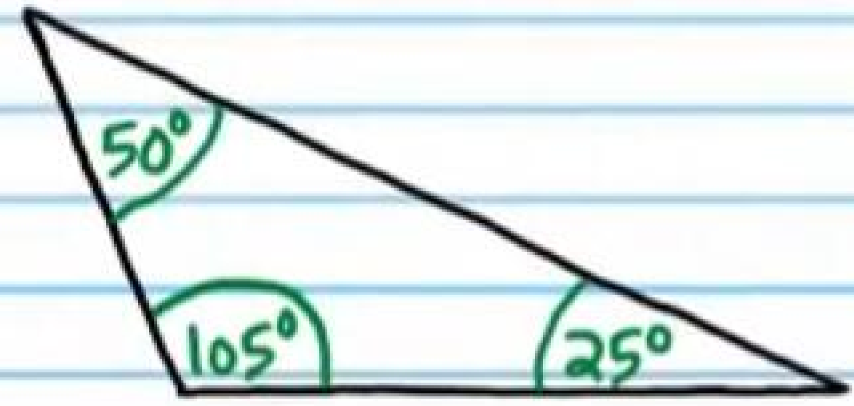
b)  Triangle

c)  Pentegon

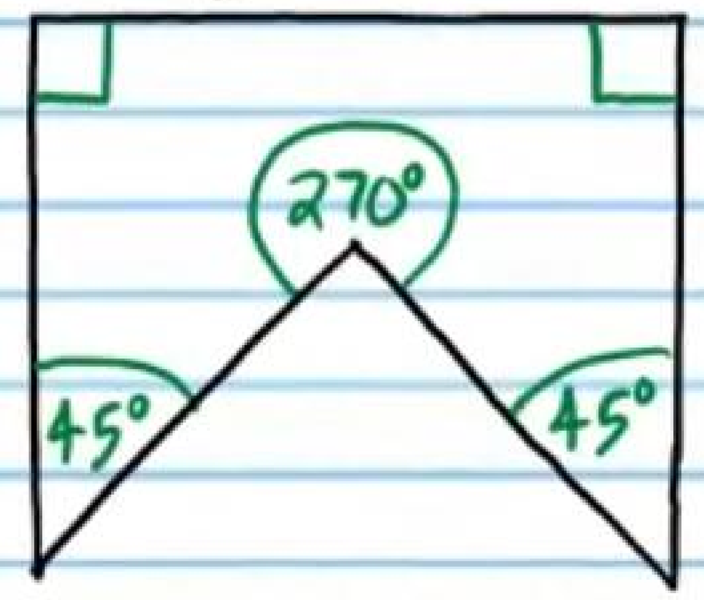
d)  Hexagon

Types of Polygons

Convex Polygon: A polygon with no interior angle measures greater than 180°.



Concave Polygon: A polygon with one or more interior angles that measure greater than 180°.

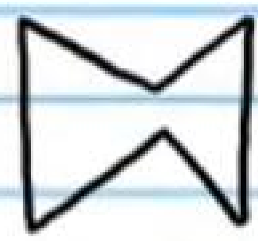


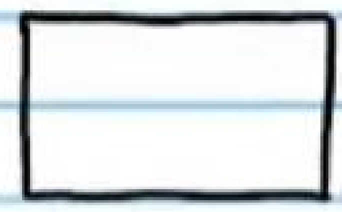
Practice

7. Label the following polygons as convex or concave.

a)  Convex

b)  Concave

c)  Concave

d)  Convex

8. Label the following polygons as convex or concave.

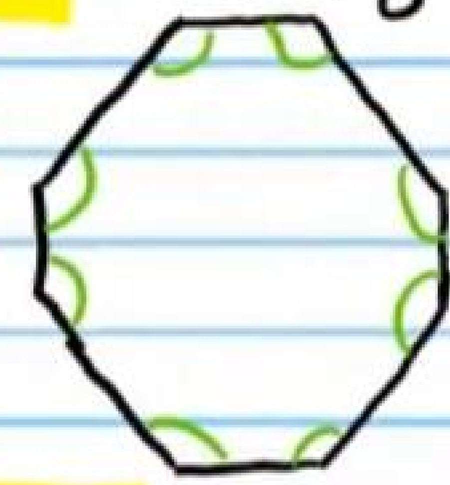
a)  Convex

b)  Concave

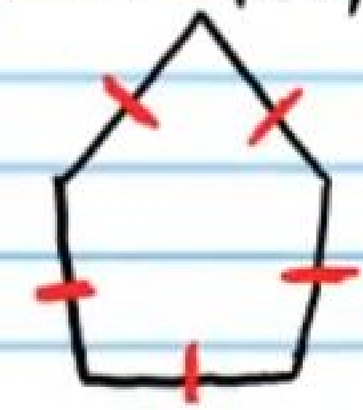
c)  Concave

d)  Convex

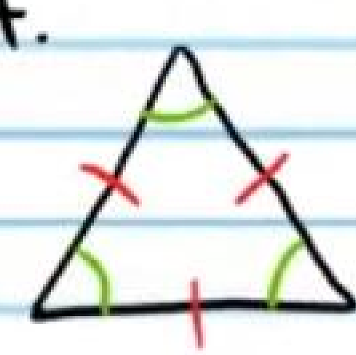
Equiangular Polygon: A polygon with all angles congruent.



Equilateral Polygon: A polygon with all sides congruent.

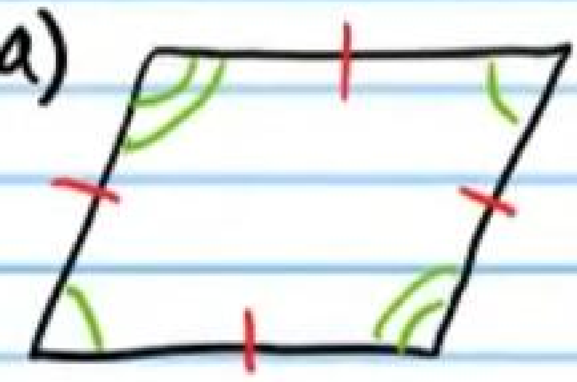


Regular Polygon: A polygon with all angles congruent and all sides congruent.

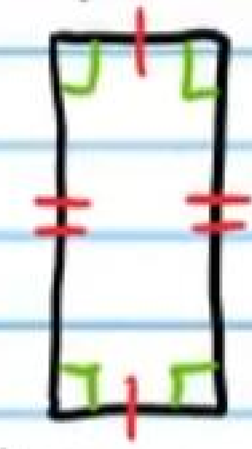


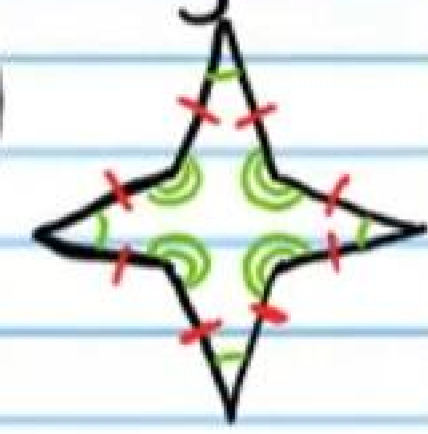
Practice

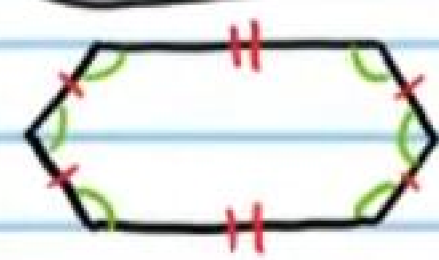
Label the following polygons as equiangular, equilateral, or regular.

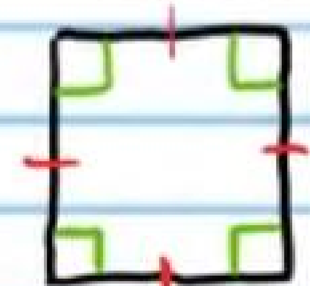
9. a)  Equilateral

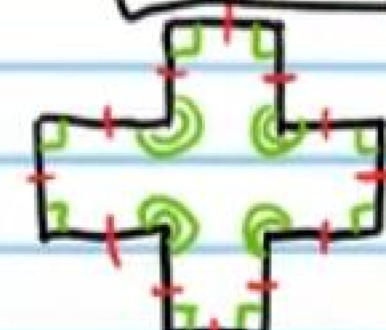
b)  Regular

c)  Equiangular

d)  Equilateral

10. a)  Equiangular

b)  Regular

c)  Equilateral

d)  Equiangular

Geometric Formulas

Perimeter of a Rectangle or Square: $P = 2L + 2W$

Area of a Rectangle or Square: $A = LW$

Area of a Triangle: $A = \frac{bh}{2}$

Volume of a Rectangular Prism: $V = LWH$

Surface Area of a Rectangular Prism: $A = 2LW + 2LH + 2WH$

Diameter of a Circle: $D = 2r$

Area of a Parallelogram: $A = bh$

Perimeter of a Circle (i.e. Circumference): $C = 2\pi r$ or πD

Area of a Circle: $A = \pi r^2$

Volume of a Sphere: $V = \frac{4\pi r^3}{3}$

Surface Area of a Sphere: $A = 4\pi r^2$

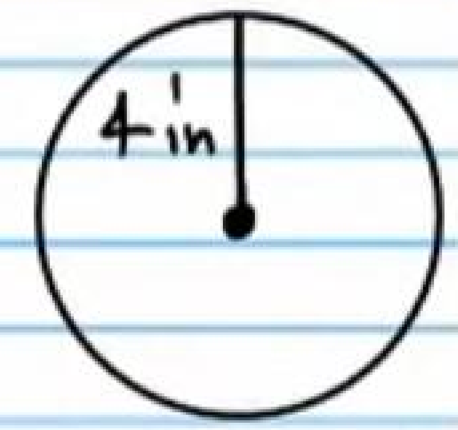
Volume of a Cylinder: $V = \pi r^2 h$

Volume of a Cone: $V = \frac{\pi r^2 h}{3}$

Area of a Trapezoid: $A = \frac{h(b_1 + b_2)}{2}$

Practice

11) Use the circle below to find the following values.



a) Diameter

$D = 2r$

$D = 2(4)$

$\boxed{D = 8 \text{ in}}$

b) Circumference

$C = 2\pi r$

$C = 2\pi(4)$

$\boxed{C = 8\pi \text{ in}}$

c) Area

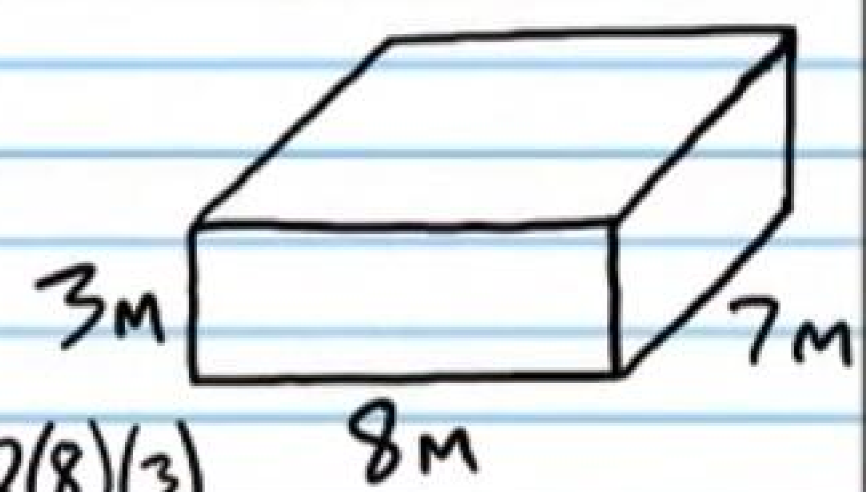
$A = \pi r^2$

$A = \pi(4)^2$

$A = \pi(16)$

$\boxed{A = 16\pi \text{ in}^2}$

12) Use the rectangular prism below to find the following values.



a) Volume

$V = LWH$

$V = (7)(8)(3)$

$\boxed{V = 168 \text{ m}^3}$

b) Surface Area

$A = 2LW + 2LH + 2WH$

$A = 2(7)(8) + 2(7)(3) + 2(8)(3)$

$A = 112 + 42 + 48$

$\boxed{A = 202 \text{ m}^2}$

(13) Find the area of the triangle below.

$$A = \frac{bh}{2}$$

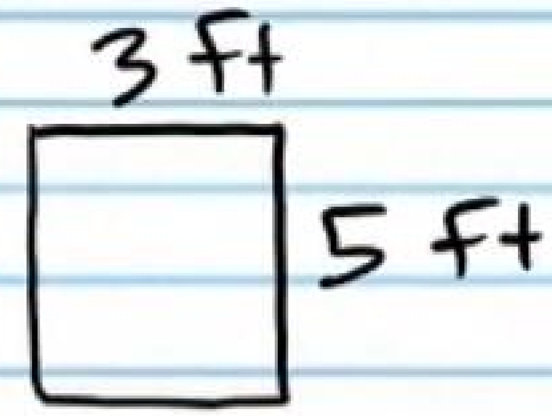
$$A = \frac{20(11)}{2}$$

$$A = \frac{220}{2}$$

$$\boxed{A = 110 \text{ km}^2}$$

More Practice

(14) Find the following values using the rectangle below.



a) Perimeter

$$P = 2W + 2L$$

$$P = 2(3) + 2(5)$$

$$P = 6 + 10$$

$$\boxed{P = 16 \text{ ft}}$$

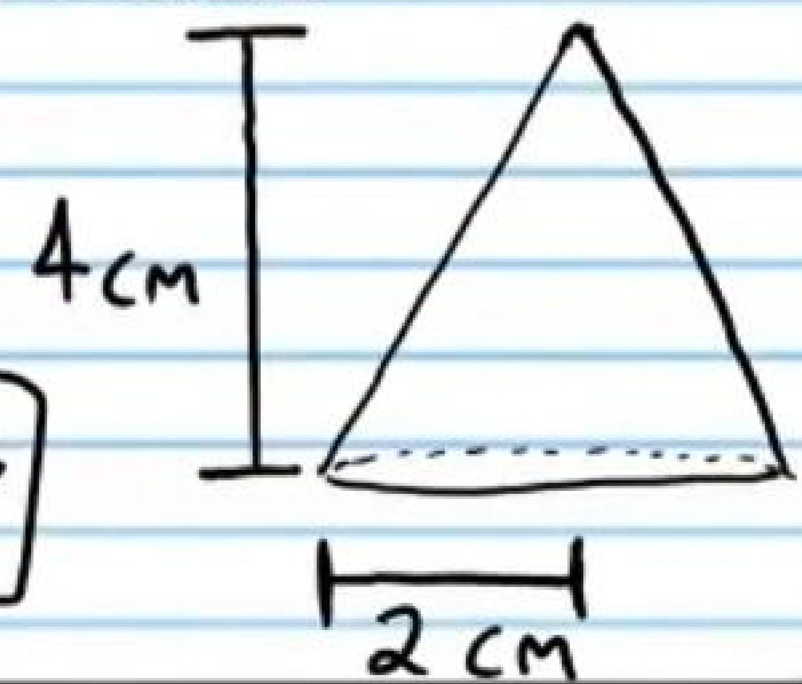
b) Area

$$A = LW$$

$$A = 5(3)$$

$$\boxed{A = 15 \text{ ft}^2}$$

(15) Find the volume of the cone below.

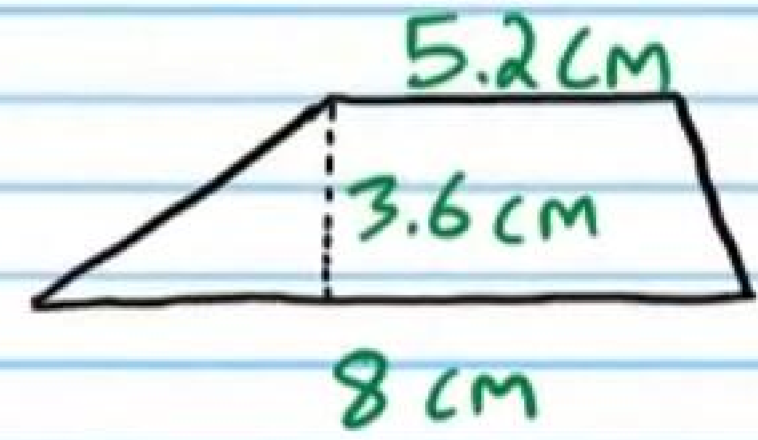


$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi(2)^2(4)$$

$$V = \frac{1}{3}\pi(4)(4) = \frac{1}{3}\pi(16) = \boxed{\frac{16}{3}\pi \text{ cm}^3}$$

(16) Find the area of the trapezoid below.



$$A = \frac{h(b_1 + b_2)}{2}$$

$$A = \frac{3.6(8 + 5.2)}{2}$$

$$A = \frac{3.6(13.2)}{2}$$

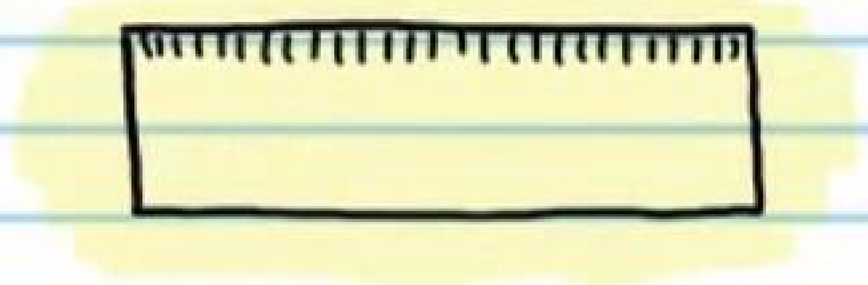
$$A = \frac{47.52}{2}$$

$$\boxed{A = 23.76 \text{ cm}^2}$$

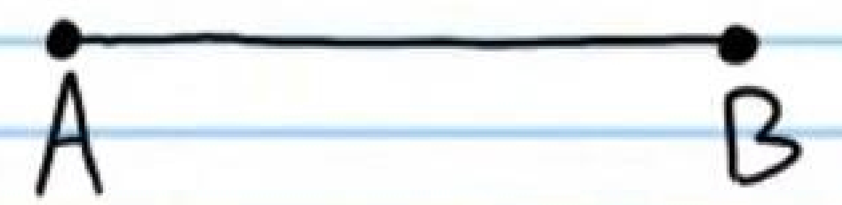
Constructions

Compass

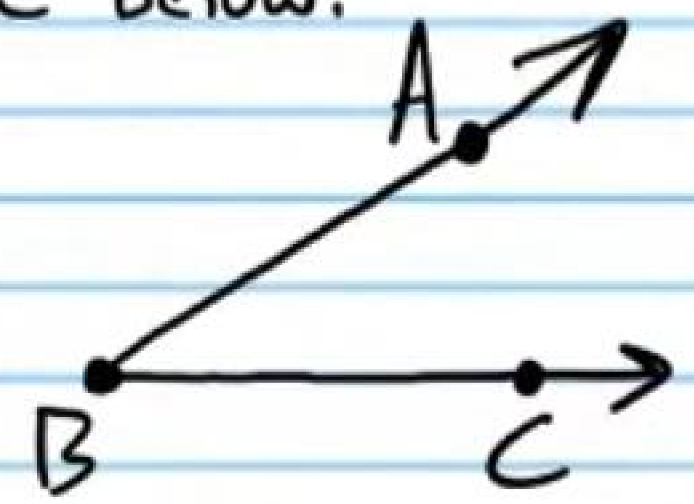
Ruler



I) Draw a segment congruent to $\overline{AB}$ below.

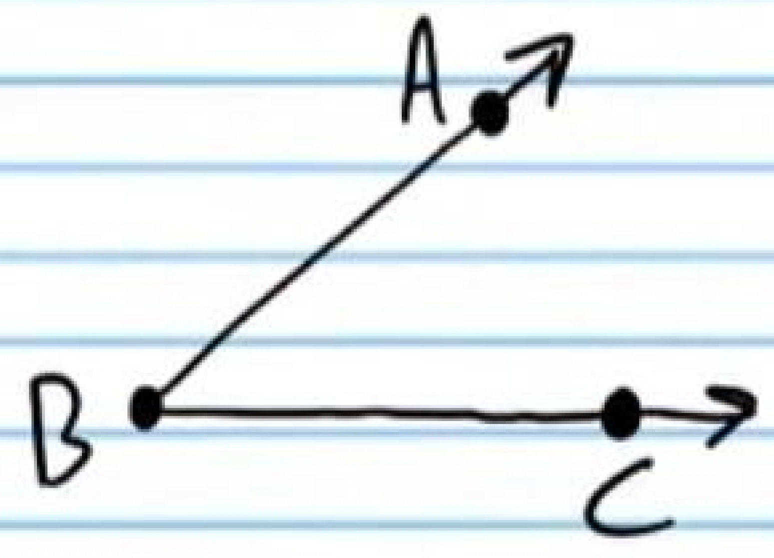


II) Draw an angle congruent to $\angle ABC$ below.



III) Draw a perpendicular segment bisector through $\overline{AB}$ below.

IV) Draw an angle bisector through $\angle ABC$ below.



Some of the constructions below use different letters than those in the problems at the left.